Mapping the Narrow Corridor with Large Language Models: A Provider-Agnostic Method for Quantifying State and Society Power Trajectories

Ebrahim M. Songhori[1]

[1]Ricursive ebrahim@ricursive.com , [affiliation — edit]

Abstract

Acemoglu and Robinson's *The Narrow Corridor* argues that liberty emerges from a balance between the power of the state and the power of society, and traces national histories as paths through a two-dimensional (state, society) space. The book supplies the conceptual space but no quantified trajectories: assigning numbers to "state power" and "society power" at each historical moment is precisely the labor-intensive, judgment-laden task that has kept the framework qualitative. We ask whether large language models (LLMs) can operationalize this task at scale. We introduce a reproducible, provider-agnostic pipeline that scores a country's trajectory period by period using chain-of-thought prompting (list events and trends first, then score) and in-context learning (prior periods anchor later ones), against an explicit rubric and returning schema-validated numeric output. Using this pipeline we produce a trajectory atlas for six countries (Iran, France, the United Kingdom, the United States, China, and Chile) and compare four models spanning three providers and both closed and open weights. We validate the LLM scores three ways: correlation with established expert indices (V-Dem, Polity5, Freedom House), inter-model and run-to-run agreement as a reliability measure, and qualitative case studies of well-documented transitions. Finally we discuss the distinct bias profiles of an LLM annotator versus a human researcher or expert panel performing the same coding task, and argue that LLMs offer scalability, reproducibility, and transparent, restatable instructions at the cost of opaque training-data bias and a tendency toward received historical consensus. Code, prompts, and all runs are released, alongside an interactive web gallery of the animated trajectories.

1 Introduction

In *The Narrow Corridor* [2], Acemoglu and Robinson propose that liberty is neither the natural product of a strong state nor of a strong society, but of a contested balance between the two. They picture a country's history as a path through a plane whose horizontal axis is the power of civil society and whose vertical axis is the power of the state; a "narrow corridor" running between the axes marks the region where the two powers grow together and liberty is sustained. The book is rich in narrative but contains *no plotted trajectories*. The obstacle is not conceptual but operational: turning centuries of a nation's history into a sequence of $(\text{society}, \text{state})$ coordinates requires an enormous amount of expert judgment, and any such coding is vulnerable to the biases, selective readings, and coarse simplifications that historians rightly distrust.

This is a measurement problem, and measurement problems in the social sciences increasingly admit machine assistance. Large language models have been shown to annotate and classify political text at or above the level of crowd workers [11], and a growing literature examines their use as instruments in computational social science [20]. We ask a narrower, concrete question: *can an LLM place a country in the Narrow Corridor space, period by period, in a way that is reproducible, defensible against external ground truth, and useful to a researcher?*

We take an applied, digital-humanities stance. Our primary artifact is a **six-country trajectory atlas** — Iran, France, the United Kingdom, the United States, China, and Chile — rendered as time-shaded paths through the corridor, together with the cross-country patterns those paths reveal. The method and its validation are the backing that makes the atlas trustworthy rather than decorative. Concretely, we contribute:

1. **A scoring method.** A period-by-period pipeline combining an anchored 0–10 rubric, chain-of-

thought elicitation of events and trends [19], and in-context anchoring on prior periods [5, 8], emitting schema-validated numeric scores.

2. **A provider-agnostic, open implementation.** Any model reachable through a unified interface can be swapped by changing one string; we release code, prompts, caches, and every run.

3. **A trajectory atlas and cross-country reading** of six countries.

4. **An interactive, publicly hosted web gallery.** A static site (GitHub Pages) presenting every trajectory as a time-shaded plot with *click-to-play animations* that reveal the path period by period and annotate the historical event driving each move; readers explore and compare models directly in the browser, turning a static figure into a navigable atlas.

5. **A three-way validation** of the scores against expert indices, against each other across models, and against documented history.

6. **A discussion of measurement bias**: what an LLM annotator gets right and wrong relative to a human researcher or an expert panel doing the identical task.

2 Related Work

The Narrow Corridor and measuring state/society power. The (state, society) framing extends the institutional program of Acemoglu and Robinson [1, 2]. Quantifying the two axes connects to decades of effort to measure regime characteristics and state capacity: the Varieties of Democracy (V-Dem) project aggregates expert codings into latent indices including a core civil-society index and measures of state administrative capacity [6, 17]; Polity5 codes regime authority on an autocracy–democracy scale [15]; and Freedom House rates political rights and civil liberties [10]. These projects are themselves large expert coding operations, which makes them both our natural validation targets (Section 5.3) and our natural comparison point when we ask what an LLM changes about the coding process (Section 6).

Text as data and LLMs as social-science instruments. The use of automated methods to extract measures from text is well established [12]. Recent work reports that instruction-tuned LLMs match or exceed human annotators on political text-labeling tasks [11] and surveys their broader promise and pitfalls for computational social science [20]. A related strand uses LLMs to *simulate* human populations [3] and debates whether they can stand in for human participants at all [7]. Our task differs from labeling or simulation: we ask the model to produce a continuous, temporally coherent *measurement* of a theoretical construct, then check that measurement against independent data.

Prompting. Our pipeline relies on two now-standard techniques. Chain-of-thought prompting [19] improves reasoning by eliciting intermediate steps; we use it to force the model to enumerate historical events and trends *before* scoring. In-context learning [5, 8] lets the model condition on examples provided at inference time; we feed all prior periods' scores forward so the trajectory is temporally coherent rather than a sequence of independent guesses. Both techniques exploit capabilities that emerge as model and corpus size grow [13].

Bias and reliability. LLMs inherit biases from training data [4] and exhibit measurable political leanings: they do not reflect a representative cross-section of opinion [18], can carry the political slant of their pretraining corpora into downstream behavior [9], and have been probed directly for partisan bias [16]. These findings motivate both our reliability analysis — borrowing the logic of inter-coder reliability from content analysis [14] — and our discussion of how LLM bias compares to the well-known biases of human expert coders.

3 Method

The space. We place a country at each time period t at a point $(\mathrm{soc}_t, \mathrm{sta}_t) \in [0, 10]^2$, where soc_t is the power of civil society and sta_t is the power of the state. A period is a fixed span of years (e.g., a decade). The corridor is drawn as two dashed boundary lines around the diagonal, following the book's picture of a balanced band.

Anchored rubric. A free-floating request for "two numbers" is not a measurement. Every scoring prompt embeds a fixed 0–10 rubric that defines each band of state power (from 0–2, collapsed/nonexistent, to 9–10, overwhelming penetration of society) and society power (from 0–2, atomized, to 9–10, society pervasively checks the state), plus explicit neutrality guidance: judge each period on its own terms, weigh evidence for and against a shift, avoid presentism, and treat the two axes as independent. The rubric is what makes scores comparable across periods, countries, and models.

Chain-of-thought: events then scores. For each period the model is first asked to list the major events and trends affecting state or society power, separating slow-moving trends from discrete events and noting the direction of each effect. This narrative is then inserted into a second prompt that requests the numeric score, so the number is conditioned on explicit reasoning rather than produced in one leap [19].

In-context learning: temporal anchoring. Starting from the earliest period (scored on absolute terms), we accumulate every prior period's $(\mathrm{soc}, \mathrm{sta})$ pair and feed the running trajectory into each subsequent scoring prompt [5]. The model reports both the change from the previous period and the resulting absolute values, which discourages the sequence from drifting and keeps a quiet period near zero change.

Structured, validated output. Each scoring call returns a JSON object — the two power values, their per-period changes, a short key event, and a one-sentence justification — validated against a fixed schema. A malformed response triggers one reminder retry and then a numeric fallback extractor before an error is raised. This replaces the brittle free-text parsing of earlier prototypes and makes the pipeline robust across providers whose JSON conventions differ.

Provider-agnostic access and caching. All model calls go through a single unified interface, so a model is selected by one identifier string and providers can be swapped without code changes. Because the pipeline makes one or two calls per period and is therefore expensive, every response is cached on disk keyed by (model, prompt); re-runs and visualization iteration are free, and the released cache makes our exact runs reproducible.

Visualization. A run renders as (i) a static plot of society vs. state power with the path time-shaded from early (dark) to late (bright) and a year colorbar, the largest moves annotated with their year and key event; and (ii) an animation that reveals the path period by period, drawing an arrow for the move each event caused and captioning it. Figure 1 is built from the static renderings.

4 Experimental Setup

Countries. We study six countries chosen to span regions of the corridor and of the world: **Iran (Persia)**, **France**, the **United Kingdom**, the **United States**, **China**, and **Chile**. Each is scored from a country-appropriate start year through 2020 in [step]-year periods (Table 1 lists windows).

Models. We compare four models spanning three provider families and both closed and open weights: `gemini-3.5-flash`, `claude-opus-4-8`, `gpt-4o`, and an open-weight `qwen` model. All are driven through the same interface with identical prompts, isolating model effects from prompt effects.

Table 1: External validation: correlation of LLM scores (`claude-opus-4-8`) with expert indices at period-midpoint years. Society vs. V-Dem core civil society; State vs. V-Dem rule-of-law proxy.

Country	Window	N	Society r	State r
Iran (Persia)	1880–2020	[–]	[–]	[–]
France	1789–2020	[–]	[–]	[–]
United Kingdom	1789–2020	[–]	[–]	[–]
United States	1789–2020	[–]	[–]	[–]
China	1880–2020	[–]	[–]	[–]
Chile	1818–2020	[–]	[–]	[–]

Validation data. For external validation we align each period's midpoint year to V-Dem [6], Polity5 [15], and Freedom House [10]. We map LLM *state power* to V-Dem state-capacity/authority measures and LLM *society power* to the V-Dem core civil-society index, and report rank and linear correlations over the overlapping years per country.

5 Results

5.1 Trajectory atlas

Figure 1 shows the six trajectories under `claude-opus-4-8`. **[One paragraph reading the atlas: which countries sit inside vs. above vs. below the corridor, and the shape of the paths (e.g., loops indicating oscillation, long diagonal runs indicating co-growth).]**

5.2 Cross-country patterns

[Report patterns visible across the atlas: e.g., whether corridor-resident countries (candidate: UK, France late) show tighter co-growth than outside-corridor countries (candidate: China high-state/low-society; pre-modern cases low-state), and how sharp transitions (revolutions, coups) appear as large single-period jumps.]

5.3 External validation

Table 1 reports correlations between LLM scores and expert indices over overlapping years. **[Summarize: society power vs. V-Dem civil society $r = \ldots$; state power vs. state-capacity $r = \ldots$; where correlation is high vs. where it breaks down.]**

`iran-persia__anthropic-claude-opus-4-8`
(panel not generated)

`france__anthropic-claude-opus-4-8`
(panel not generated)

`united-kingdom__anthropic-claude-opus-4-8`
(panel not generated)

`united-states__anthropic-claude-opus-4-8`
(panel not generated)

`china__anthropic-claude-opus-4-8`
(panel not generated)

`chile__anthropic-claude-opus-4-8`
(panel not generated)

Figure 1: Trajectory atlas: society power (x) vs. state power (y) for six countries under `claude-opus-4-8`. Paths are time-shaded (dark = early, bright = late); dashed lines mark the corridor. **[Regenerate with `paper/experiments/run_experiments.py`.]**

Table 2: Inter-model agreement: mean pairwise Pearson r between models on the same country, over shared periods. Higher = more reliable.

Country	#models	Society r	State r
Iran (Persia)	0	[–]	[–]
France	0	[–]	[–]
United Kingdom	0	[–]	[–]
United States	0	[–]	[–]
China	0	[–]	[–]
Chile	0	[–]	[–]
Overall	–	[–]	[–]

5.4 Model comparison and reliability

Treating each model as a "rater," Table 2 reports pairwise agreement between models on the same country, and (where repeated runs are enabled) run-to-run variance. **[Summarize: which axis is more stable, which model pairs agree most, and whether the open-weight model diverges from the closed frontier models.]**

5.5 Qualitative case studies

[Two or three short reads checking that trajectories match documented history: Iran around 1979 (sharp state consolidation / society suppression after the revolution); Chile 1973 and 1990 (coup and redemocratization); and a third (e.g., UK gradual co-growth, or China post-1949). Note where the model matches consensus and where it is anachronistic or flattens contested episodes.]

6 Discussion: LLM vs. Human Researcher or Expert Panel

Both an LLM and a human expert (or panel) performing this coding are biased; the useful question is *how the bias profiles differ*, and which task properties favor which coder.

What a human panel brings. Domain expertise, sensitivity to contested historiography, the ability to flag "this case is genuinely disputed," and accountable, attributable judgment. Established indices such as V-Dem manage individual coder bias with multiple experts per case and a latent measurement model that estimates and corrects for coder disagreement [17]; content analysis formalizes this as inter-coder reliability [14]. The costs are equally real: expert panels are slow, expensive, hard to scale to arbitrary country-periods, and carry their own systematic biases — national, ideological, disciplinary — that are difficult to audit because they live in individual judgment.

What an LLM changes. An LLM annotator is fast, cheap at scale, and *transparent in its instructions*: the rubric and prompts are explicit artifacts that can be inspected, criticized, versioned, and re-run identically. Its bias, however, is *opaque in origin*: it reflects the distribution of its training data [4, 9, 18], which over-represents English-language, recent, and Western sources, and it tends toward received consensus, likely flattening contested or under-documented episodes rather than flagging them. It has no accountable author, and its scores can shift with model version or provider (Section 5.4).

Reliability as the bridge. The inter-coder logic that human panels use for humans applies to models: agreement across independent models, and stability across repeated runs, is a measurable proxy for reliability. High inter-model agreement plus high correlation with expert indices is our evidence that a score reflects a shared signal rather than one model's idiosyncrasy; disagreement usefully localizes exactly the contested cases a human panel would also flag.

When to use which. We argue the LLM pipeline is best positioned as a *scalable first pass and a reproducible baseline* — generating candidate trajectories over many countries cheaply, surfacing where models disagree, and freeing expert attention for the contested cases — rather than as a replacement for expert coding where accountability and contested-case judgment are paramount.

7 Limitations

The two axes compress vast, contested constructs into single numbers; the rubric mitigates but cannot eliminate this. Training-data cutoffs and coverage bias the scores toward well-documented, recent, Western-adjacent histories. We validate against expert indices that are themselves imperfect and only partially aligned in construct with our axes. Scores can depend on model version, decoding randomness, and prompt phrasing; we quantify the first two but a full prompt- sensitivity study is future work. Finally, we study six countries and four models — enough to demonstrate the method and expose model effects, not to make global claims.

8 Conclusion

We showed that LLMs can operationalize the Narrow Corridor's (state, society) space into reproducible, externally validatable trajectories, and released a provider-agnostic tool that makes doing so a one-command exercise. The resulting six-country atlas is a first quantified rendering of a framework that has been purely qualitative, and the accompanying validation and bias analysis frame honestly what such LLM-generated measurements are — and are not — good for.

Reproducibility

All code, prompts, the on-disk response cache, and every run (JSON, static plots, animations) are released at `https://github.com/esonghori/visualize_narrow_corridor_with_ai`. The exact experiments in this paper are reproduced by `paper/experiments/run_experiments.py` and `analysis.py`; see `paper/README.md`. The interactive gallery of animated trajectories is hosted at `https://esonghori.github.io/visualize_narrow_corridor_with_ai/`.

References

[1] Daron Acemoglu and James A. Robinson. *Why Nations Fail: The Origins of Power, Prosperity, and Poverty*. Crown Business, New York, 2012.

[2] Daron Acemoglu and James A. Robinson. *The Narrow Corridor: States, Societies, and the Fate of Liberty*. Penguin Press, New York, 2019.

[3] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.

[4] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pages 610–623, 2021.

[5] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, et al. Language models are few-shot learners. *Advances in Neural Information Processing Systems*, 33:1877–1901, 2020.

[6] Michael Coppedge, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, et al. V-dem codebook v14. *Varieties of Democracy (V-Dem) Project*, 2024.

[7] Danica Dillion, Niket Tandon, Yuling Gu, and Kurt Gray. Can ai language models replace human participants? *Trends in Cognitive Sciences*, 27(7): 597–600, 2023.

[8] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Jingyuan Ma, Rui Li, Heming Xia, et al. A survey on in-context learning. *Proceedings of EMNLP*, 2024.

[9] Shangbin Feng, Chan Young Park, Yuhan Liu, and Yulia Tsvetkov. From pretraining data to language models to downstream tasks: Tracking the trails of political biases leading to unfair nlp models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, 2023.

[10] Freedom House. Freedom in the world 2023: Marking 50 years in the struggle for democracy. Technical report, Freedom House, 2023.

[11] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. Chatgpt outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30), 2023.

[12] Justin Grimmer and Brandon M. Stewart. Text as data: The promise and pitfalls of automatic content analysis methods for political texts. *Political Analysis*, 21(3):267–297, 2013.

[13] Jared Kaplan, Sam McCandlish, Tom Henighan, Tom B. Brown, Benjamin Chess, Rewon Child, Scott Gray, Alec Radford, Jeffrey Wu, and Dario Amodei. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.

[14] Klaus Krippendorff. *Content Analysis: An Introduction to Its Methodology*. SAGE Publications, 4th edition, 2018.

[15] Monty G. Marshall and Ted Robert Gurr. Polity5: Political regime characteristics and transitions, 1800–2018. Technical report, Center for Systemic Peace, 2020.

[16] Fabio Motoki, Valdemar Pinho Neto, and Victor Rodrigues. More human than human: Measuring chatgpt political bias. *Public Choice*, 198:3–23, 2024.

[17] Daniel Pemstein, Kyle L. Marquardt, Eitan Tzelgov, Yi-ting Wang, Joshua Krusell, and Farhad Miri. The v-dem measurement model: Latent variable analysis for cross-national and cross-temporal expert-coded data. *V-Dem Working Paper 21, University of Gothenburg*, 2018.

[18] Shibani Santurkar, Esin Durmus, Faisal Ladhak, Cinoo Lee, Percy Liang, and Tatsunori Hashimoto. Whose opinions do language models reflect? In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023.

[19] Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc V. Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.

[20] Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024.